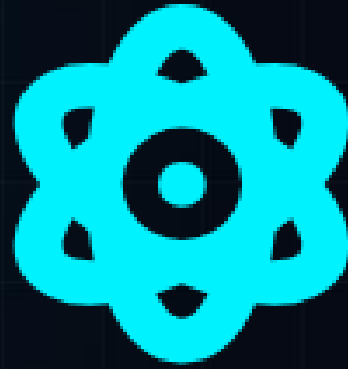
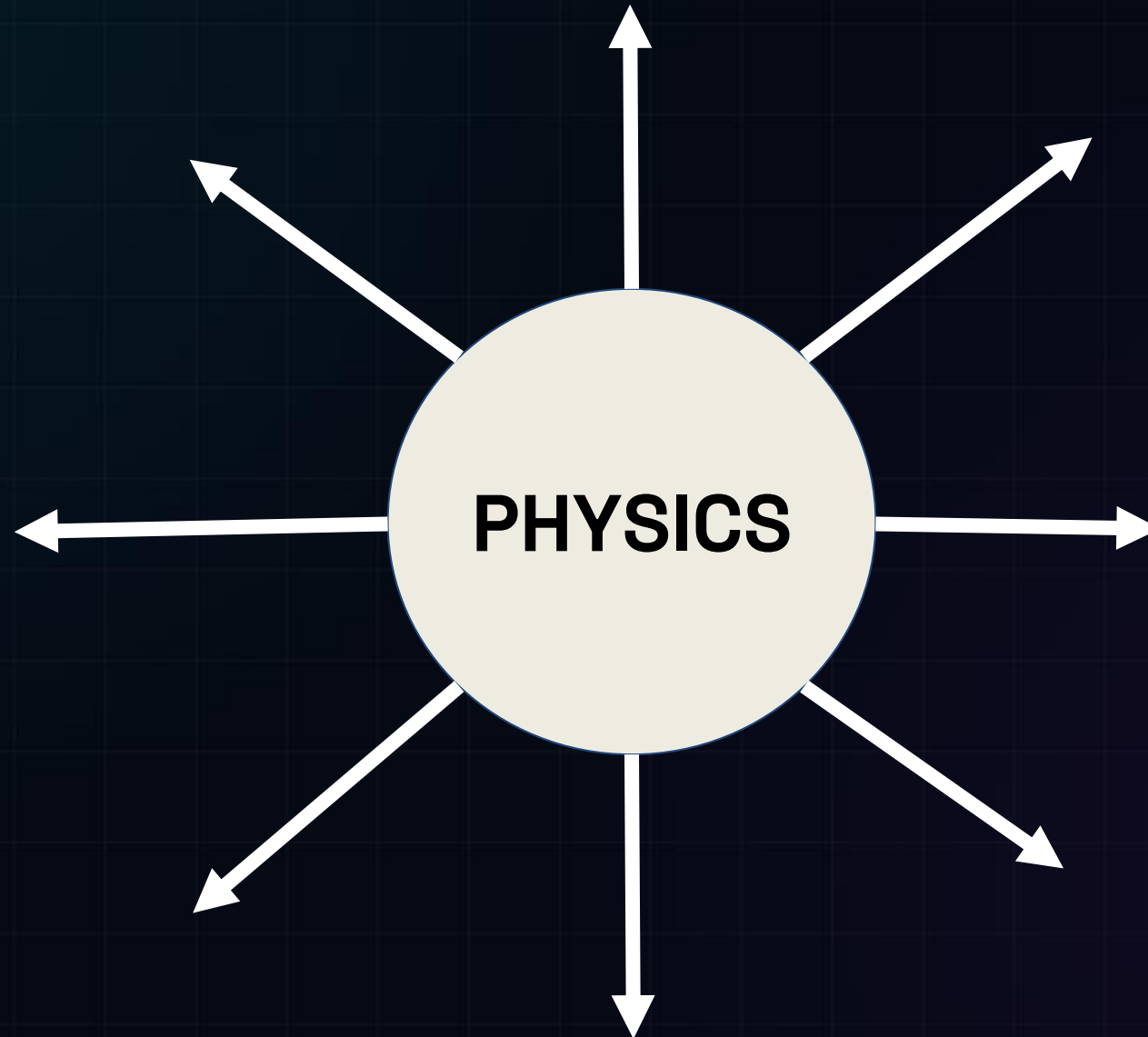




PHYSICS IN DAILY LIFE

General Science 11 | Strategic Learning Lesson

MOTIVATION: PHYSICS HOTSPOT



| MOTIVATION: PHYSICS & YOU



ACTIVITY 1

Short Video Clip: Impact of physics on medicine, transport, and communication.



ACTIVITY 2

Guided Discussion: How do these applications make our lives safer or easier?



GOAL

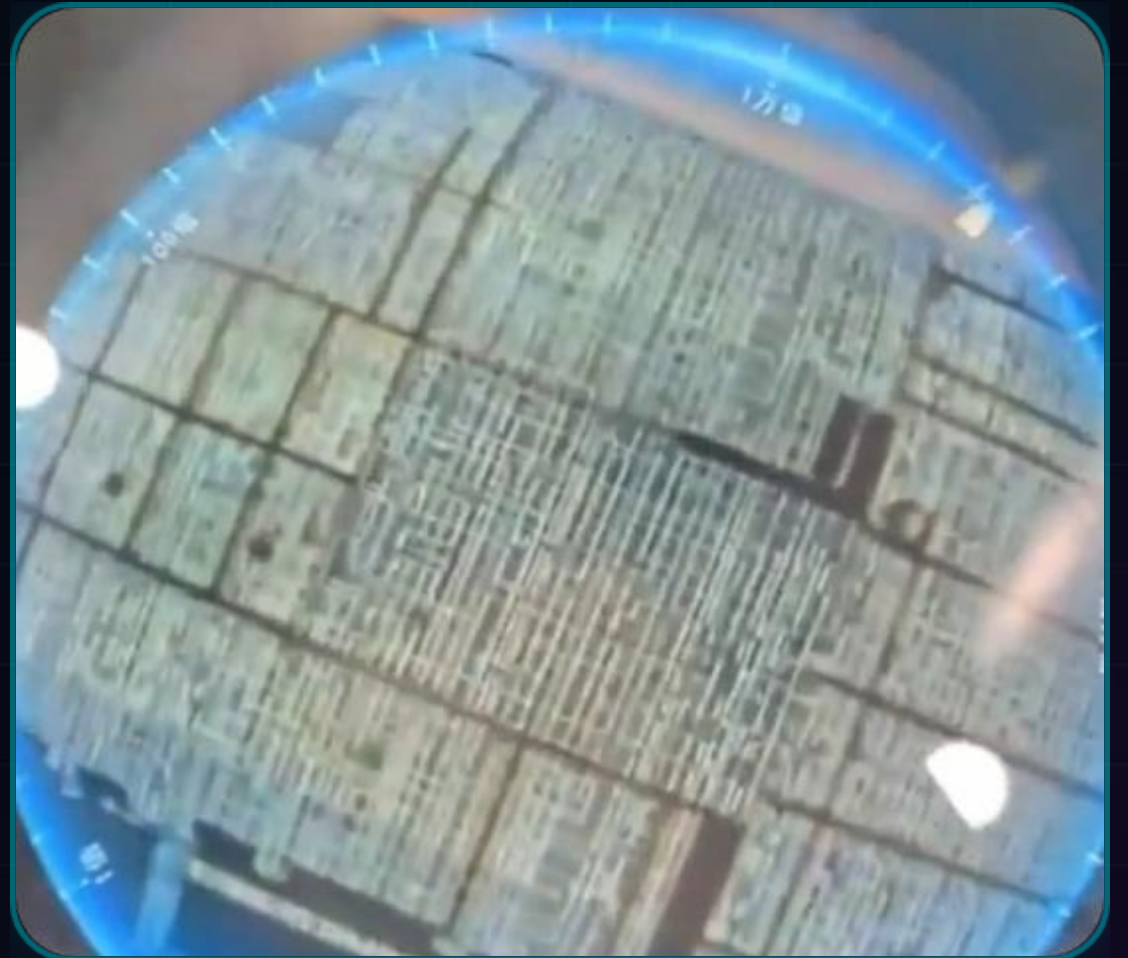
Identify the hidden ways physics enhances our quality of life across all areas.

| VOCABULARY: INNOVATION

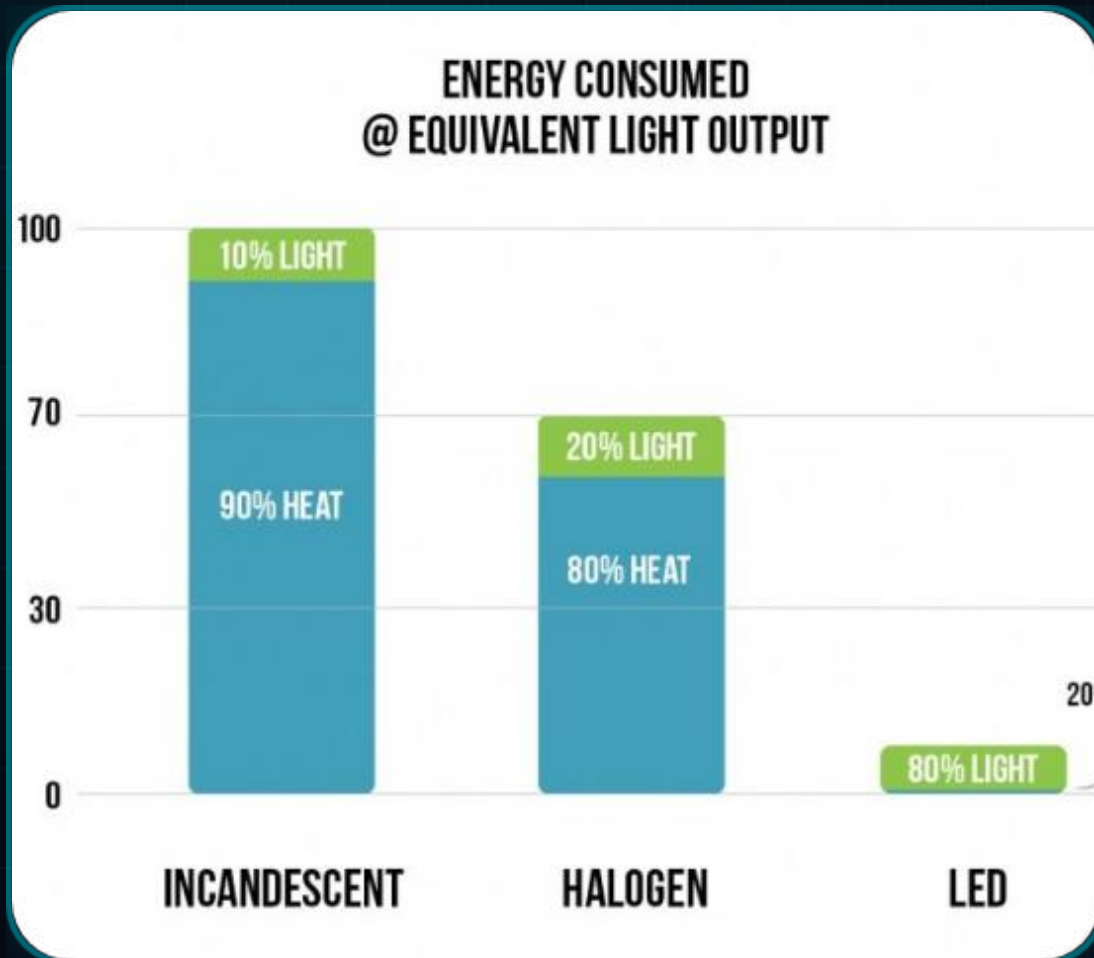
INNOVATION

Definition: Creating new methods or products that solve problems using physical laws.

Real Life: High-speed fiber optics use the physics of light to connect the world.



VOCAB: EFFICIENCY



EFFICIENCY

Definition: Achieving maximum work with minimum waste of energy or resources.

Real Life: LED bulbs save energy by producing light without the heat waste of old bulbs.

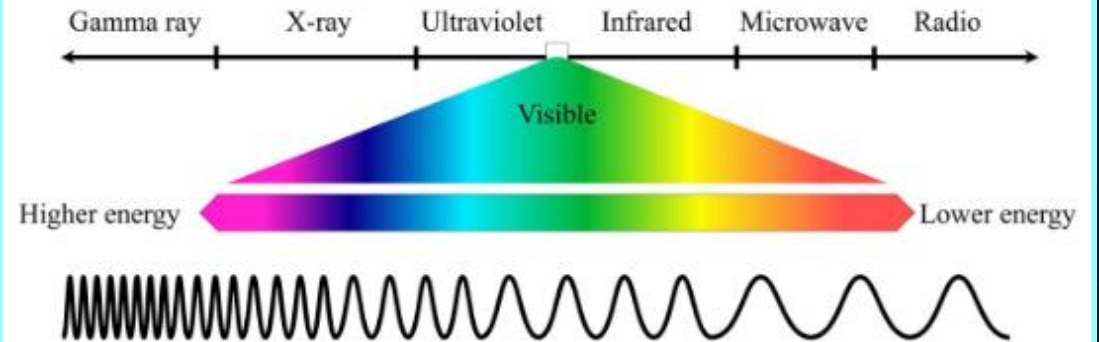
| VOCAB: EM RADIATION

ELECTROMAGNETIC RADIATION

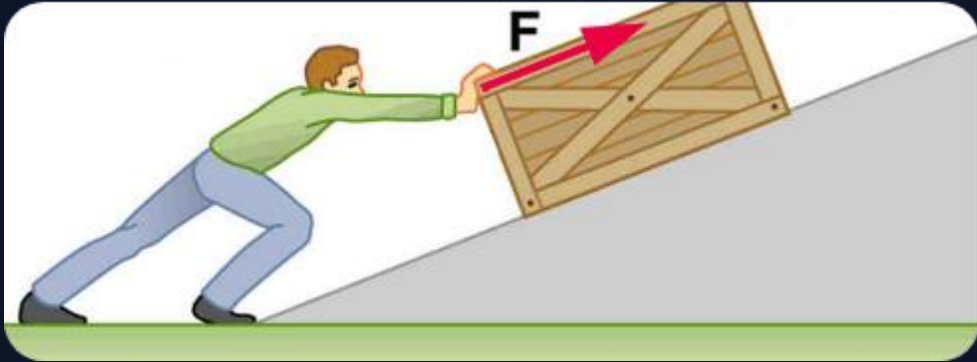
Definition: Invisible waves of energy that carry information and heat.

Real Life: From your microwave oven to your phone's 5G signal.

Electromagnetic Spectrum



| VOCAB: FORCE & ENERGY



FORCE

A push or pull that makes something move, stop, or change direction.



ENERGY

The "power" needed to do work or make something happen.

THE PHYSICS ADVANTAGE



Physics is fundamental to many modern technologies. We call it the **Advantage** because it allows us to do things impossible by nature alone—like flying or seeing through walls.

1. HOUSEHOLD ACTIVITIES

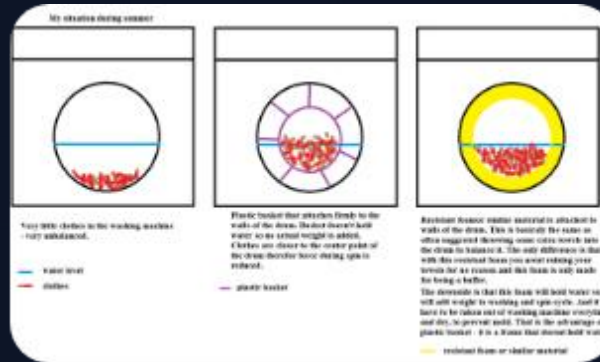
Managing energy and work at
home

| ENERGY IN THE HOME



SOLAR PANELS

Convert photons (light) into electricity via the **Photoelectric Effect**.



WASHING MACHINE

Uses **Centripetal Force** to spin water out of clothes at high speeds.



CIRCUIT BREAKERS

Uses **Electromagnetism** to cut power instantly to prevent house fires.

2. HEALTH AND SAFETY

Physics saving lives every
day

| MEDICAL PHYSICS & SAFETY

DIAGNOSTICS

MRI: Uses Magnetic Resonance to image soft tissue without surgery.

CAR SAFETY

Airbags: Uses Gas Laws and Inertia to provide an cushioned stop during a crash.



3. WORK PRODUCTIVITY

Automating the world around

US

| INDUSTRIAL ADVANTAGE



 **Robotics:** Mechanical arms use torque and levers to build cars.

 **Smartphones:** Use solid-state physics for storage and EM waves for calls.

4. LEISURE AND RECREATION

Enjoying the science of motion and
light

| SCIENCE OF FUN

SPORTS

Basketball: Projectile motion helps players calculate the perfect arc to score.

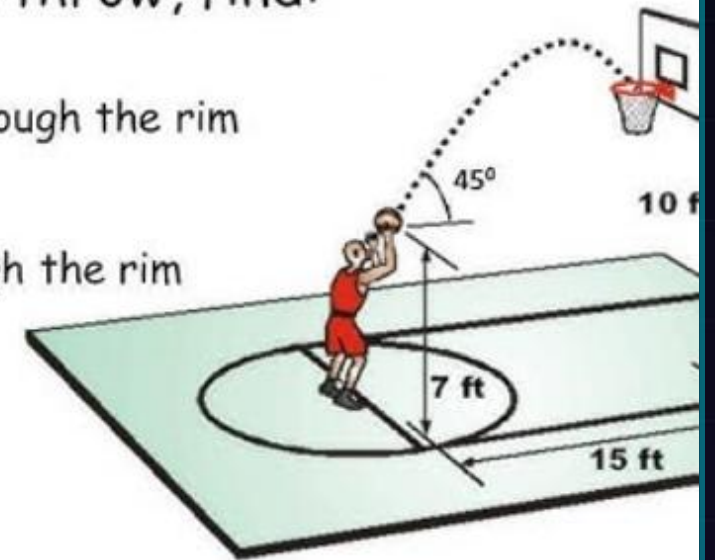
ENTERTAINMENT

Music: Wave physics allows speakers to vibrate air into beautiful sound.

Basketball Projectile Motion

A player shoots a free throw, find:

initial velocity v_0
the ball passes through the rim
maximum height
time of the ball and
time as it passes through the rim



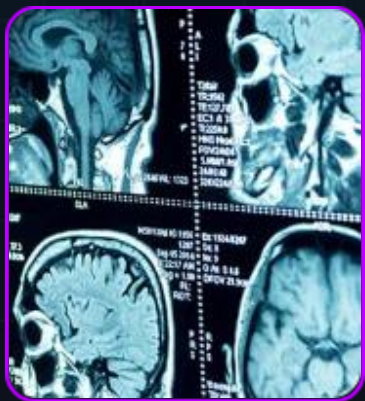
PHYSICS **NINJA**

| ASSIGNMENT BY GROUP



GROUP 1: HOUSEHOLD

Research and present how physics is applied in common household appliances and energy-saving technologies (e.g., refrigerators, solar panels, washing machines).



GROUP 2: HEALTH & SAFETY

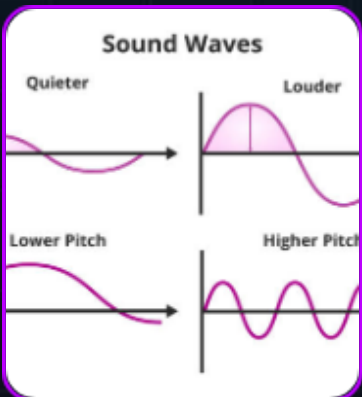
Investigate and present how physics is used in medical imaging (X-rays, MRI), safety devices (seatbelts, airbags), and public health (sanitation).

| ASSIGNMENT BY GROUP



GROUP 3: PRODUCTIVITY

Explore how physics contributes to automation (robotics), communication (internet, smartphones), and transportation (aerodynamics).



GROUP 4: LEISURE

Discover how physics enhances sports (projectile motion, aerodynamics) and entertainment (sound waves, optics in displays).

DISCUSSION & QUESTIONS

Physics isn't just in the classroom—it's in every breath you take and every tool you use.

Ready to present your group
findings?

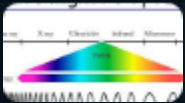
IMAGE SOURCES



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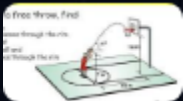
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IMAGE SOURCES



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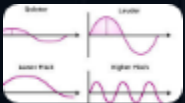
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